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Array substrate and manufacturing method thereof, and display panel

Abstract

Provided is an array substrate. The array substrate includes a display region and a non-display region located at a periphery of the display region, wherein the array substrate includes a substrate; a first transistor and a second transistor that are disposed on the substrate, wherein the first transistor is disposed in the display region, and the second transistor is disposed in the non-display region; and a data line and a pixel electrode that are disposed in the display region, wherein the data line is disposed on a side of the first active layer close to the substrate and is lapped with the first active layer, and the pixel electrode is disposed on a side of the first gate facing away from the substrate and is lapped with the first active layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. national stage of international application No. PCT/CN2022/121475, filed on Sep. 26, 2022, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the field of display technologies, and in particular relates to an array substrate and a manufacturing method thereof, and a display panel.

BACKGROUND

(3) Liquid crystal display panels have the characteristics of small size, low power consumption, no radiation and the like, and dominate the current display market.

SUMMARY

(4) Embodiments of the present disclosure provide an array substrate and a manufacturing method thereof, and a display panel. The technical solutions are as follows.

(5) In some embodiments of the present disclosure, an array substrate is provided. The array substrate includes a display region and a non-display region located at a periphery of the display region. The array substrate includes: a substrate; a first transistor and a second transistor that are disposed on the substrate, wherein the first transistor is disposed in the display region, and the second transistor is disposed in the non-display region; the first transistor includes a first gate and a first active layer, the first gate being disposed on a side of the first active layer facing away from the substrate; and the second transistor includes a second active layer and a second gate; and a data line and a pixel electrode that are disposed in the display region, wherein the data line is disposed on a side of the first active layer close to the substrate and is lapped with the first active layer, and the pixel electrode is disposed on a side of the first gate facing away from the substrate and is lapped with the first active layer; wherein the data line and the second gate are disposed in a same layer and made of a same material.

(6) In some embodiments, the first active layer includes a first conductor portion and a second conductor portion opposite to each other, and a first channel region disposed between the first conductor portion and the second conductor portion, an orthographic projection of the first channel region on the substrate is within an orthographic projection of the first gate on the substrate; wherein

(7) an orthographic projection of the data line on the substrate has an overlapping region with an orthographic projection of the first conductor portion on the substrate, and the first conductor portion is lapped with the data line; and

(8) an orthographic projection of the pixel electrode on the substrate has an overlapping region with an orthographic projection of the second conductor portion on the substrate, and the pixel electrode is lapped with the second conductor portion.

(9) In some embodiments, the array substrate further includes: a transparent connecting electrode disposed between the pixel electrode and the first active layer, wherein an orthographic projection of the transparent connecting electrode on the substrate has an overlapping region with the orthographic projection of the pixel electrode on the substrate, and has an overlapping region with the orthographic projection of the second conductor portion on the substrate, and the pixel

electrode is lapped with the second conductor portion by means of the transparent connecting electrode.

(10) In some embodiments, the array substrate further includes: a light shielding layer, wherein the light shielding layer and the data line are disposed in a same layer and made of a same material, and the orthographic projection of the first channel region on the substrate is within an orthographic projection of the light shielding layer on the substrate.

(11) In some embodiments, a plurality of first transistors are provided, and the light shielding layer includes a plurality of light shielding blocks in one-to-one correspondence with the plurality of first transistors, wherein the orthographic projection of the first channel region of the first active layer in the first transistor on the substrate is within an orthographic projection of the corresponding light shielding block on the substrate.

(12) In some embodiments, the first active layer and the second active layer are disposed in different layers, and the second active layer is closer to the substrate than the second gate is.

(13) In some embodiments, the second transistor further includes: a source and a drain which are lapped with the second active layer, wherein the source and the drain are disposed in a same layer and made of a same material, and the source and the drain are both disposed on a side of the second gate facing away from the substrate; wherein the source, the drain and the first active layer are disposed in the same layer, and a material of the first active layer is different from a material of the source and the drain; or the source, the drain and the first gate are disposed in the same layer and made of the same material.

(14) In some embodiments, in a case that the source, the drain and the first active layer are disposed in the same layer, the array substrate further includes: a first gate insulating layer disposed between the second gate and the second active layer, and an auxiliary insulating layer disposed between the first active layer and the data line; the array substrate is provided with a first via hole penetrating through the auxiliary insulating layer, and a plurality of lap joint via holes penetrating through the auxiliary insulating layer and the first gate insulating layer, wherein the first active layer is lapped with the data line through the first via hole, the source is lapped with the second active layer through one of the lap joint via holes, and the drain is lapped with the second active layer through another one of the lap joint via holes; wherein an opening of the first via hole facing away from the substrate and openings of the lap joint via holes facing away from the substrate are all on a surface of the auxiliary insulating layer facing away from the substrate.

(15) In some embodiments, the array substrate further includes: a second gate insulating layer and an interlayer dielectric layer; wherein the second active layer is disposed on a side of the first gate insulating layer close to the substrate; the data line and the second gate are both disposed between the first gate insulating layer and the auxiliary insulating layer; the first active layer, the source and the drain are all disposed between the auxiliary insulating layer and the second gate insulating layer; and the first gate is disposed between the second gate insulating layer and the interlayer dielectric layer.

(16) In some embodiments, the first active layer and the second active layer are disposed in a same layer but made of different materials, and the second gate is closer to the substrate than the second active layer is.

(17) In some embodiments, the second transistor further includes: a source and a drain which are lapped with the second active layer, wherein the source and the drain are disposed in a same layer and made of a same material, and the source and the drain are both disposed on a side of the second active layer facing away from the substrate; wherein the array substrate further includes: a connecting electrode, wherein the connecting electrode is lapped with the first active layer and the data line, and the connecting electrode, the first gate, the source and the drain are disposed in a same layer and made of a same material.

(18) In some embodiments, the array substrate further includes: a first gate insulating layer disposed between the second gate and the second active layer, and a second gate insulating layer

disposed between the first active layer and the first gate; wherein the array substrate is provided with a first sub-via hole penetrating through the second gate insulating layer and the first gate insulating layer, and a second sub-via hole and a plurality of lap joint via holes penetrating through the second gate insulating layer, wherein the connecting electrode is lapped with the first active layer through the second sub-via hole, and is lapped with the data line through the first sub-via hole, the source is lapped with the second active layer through one of the lap joint via holes, and the drain is lapped with the second active layer through another one of the lap joint via holes; wherein an opening of the first sub-via hole facing away from the substrate, an opening of the second sub-via hole facing away from the substrate, and openings of the lap joint via holes facing away from the substrate are all on a surface of the second gate insulating layer facing away from the substrate.

(19) In some embodiments, the array substrate further includes: an interlayer dielectric layer; wherein the second gate and the data line are disposed on a side of the first gate insulating layer close to the substrate; the first active layer and the second active layer are disposed between the first gate insulating layer and the second gate insulating layer; and the source, the drain, the connecting electrode and the first gate are all disposed between the second gate insulating layer and the interlayer dielectric layer.

(20) In some embodiments, the array substrate further includes: a transparent connecting electrode disposed on a side of the interlayer dielectric layer facing away from the substrate, and a planarization layer disposed on a side of the transparent connecting electrode facing away from the substrate, the pixel electrode being disposed on a side of the planarization layer facing away from the substrate; wherein the array substrate is provided with a connecting via hole penetrating through the planarization layer, and a second via hole penetrating through the interlayer dielectric layer and the second gate insulating layer, the pixel electrode is lapped with the transparent connecting electrode through the connecting via hole, and the transparent connecting electrode is lapped with the first active layer through the second via hole.

(21) In some embodiments, the array substrate further includes: a passivation layer disposed on a side of the pixel electrode facing away from the substrate, and a common electrode layer disposed on a side of the passivation layer facing away from the substrate.

(22) In some embodiments, the common electrode layer includes: a light shielding portion and a transparent conductive portion disposed on a side of the light shielding portion facing away from the substrate, wherein an orthographic projection of the light shielding portion on the substrate is not overlapped with an orthographic projection of the pixel electrode on the substrate.

(23) In some embodiments, the array substrate further includes: a support part, wherein a portion of the support part is in the connecting via hole, and the other portion is outside the connecting via hole.

(24) In some embodiments, the first transistor is an oxide thin film transistor, and the second transistor is a low temperature poly-silicon thin film transistor.

(25) In some embodiments, the array substrate further includes: a gate drive circuit disposed in the non-display region, wherein the gate drive circuit includes a plurality of second transistors.

(26) In some embodiments of the present disclosure, a method of manufacturing an array substrate is provided. The array substrate includes a display region and a non-display region located at a periphery of the display region. The method includes: forming a first transistor, a second transistor, a data line and a pixel electrode on a substrate; wherein the first transistor is disposed in the display region, and the second transistor is disposed in the non-display region; the first transistor includes a first gate and a first active layer, the first gate being disposed on a side of the first active layer facing away from the substrate; and the second transistor includes a second active layer and a second gate; the data line and the pixel electrode are both disposed in the display region, the data line is disposed on a side of the first active layer close to the substrate, the data line is lapped with the first active layer, the pixel electrode is disposed on a side of the first gate facing away from the

substrate, and the pixel electrode is lapped with the first active layer; wherein the data line and the second gate are disposed in a same layer and made of a same material.

(27) In some embodiments of the present disclosure, a display panel is provided. The display panel includes: an array substrate and a color filter substrate disposed opposite to each other, and a liquid crystal layer disposed between the array substrate and the color filter substrate, wherein the array substrate is the array substrate as described above.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) To describe the technical solutions in the embodiments of the present disclosure more clearly, the following briefly introduces the accompanying drawings required for describing the embodiments. The accompanying drawings in the following descriptions show merely some embodiments of the present disclosure, and persons of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative effort.

(2) FIG. 1 is a top view of an array substrate according to some embodiments of the present disclosure;

(3) FIG. 2 is a schematic structural diagram of film layers in the array substrate shown in FIG. 1 along a line A-A';

(4) FIG. 3 is a schematic structural diagram of film layers in another array substrate according to some embodiments of the present disclosure;

(5) FIG. 4 is a schematic structural diagram of film layers in still another array substrate according to some embodiments of the present disclosure;

(6) FIG. 5 is a schematic structural diagram of film layers in still another array substrate according to some embodiments of the present disclosure;

(7) FIG. 6 is a schematic structural diagram of film layers in an array substrate according to some embodiments of the present disclosure;

(8) FIG. 7 is a schematic structural diagram of film layers in another array substrate according to some embodiments of the present disclosure;

(9) FIG. 8 shows a method of manufacturing an array substrate according to some embodiments of the present disclosure;

(10) FIG. 9 is a schematic diagram of formation of a second active layer on a substrate according to some embodiments of the present disclosure;

(11) FIG. 10 is a schematic diagram of formation of a first gate insulating layer and a first conductive layer sequentially on a second active layer according to some embodiments of the present disclosure;

(12) FIG. 11 is a schematic diagram of formation of an auxiliary insulating layer on a first conductive layer according to some embodiments of the present disclosure;

(13) FIG. 12 is a schematic diagram of formation of a second conductive layer according to some embodiments of the present disclosure;

(14) FIG. 13 is a schematic diagram of formation of a first active layer according to some embodiments of the present disclosure;

(15) FIG. 14 shows a method of manufacturing an array substrate according to some embodiments of the present disclosure;

(16) FIG. 15 is a schematic diagram of formation of a first conductive layer according to some embodiments of the present disclosure;

(17) FIG. 16 is a schematic diagram of formation of a first gate insulating layer on a first conductive layer according to some embodiments of the present disclosure;

(18) FIG. 17 is a schematic diagram of formation of a second active layer on a first gate insulating

layer according to some embodiments of the present disclosure;

(19) FIG. **18** is a schematic diagram of formation of a first active layer on a first gate insulating layer according to some embodiments of the present disclosure;

(20) FIG. **19** is a schematic diagram of formation of a second gate insulating layer on a first active layer and a second active layer according to some embodiments of the present disclosure; and

(21) FIG. **20** is a schematic diagram of formation of a second conductive layer on a second gate insulating layer according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(22) To make the objects, technical solutions, and advantages of the present disclosure clearer, the embodiments of the present disclosure are further described in detail hereinafter with reference to the accompanying drawings.

(23) Generally, a liquid crystal display panel includes an array substrate and a color filter substrate disposed opposite to each other, and a liquid crystal layer disposed therebetween. The array substrate may be integrated with a plurality of thin film transistors (TFTs) in the display region, and a plurality of data lines electrically connected to the plurality of TFTs in one-to-one correspondence, and the array substrate may be integrated with a gate driver on array (GOA) circuit in the non-display region. The GOA circuit may be electrically connected to the gate of the TFT through a gate line, so that the GOA circuit can control the TFT to be turned on and off. When the TFT is turned on, the data line electrically connected to the TFT can apply pixel voltage to a pixel electrode electrically connected to the TFT.

(24) However, the parasitic capacitance between the pixel electrode and the data line in the current array substrate is large, resulting in a high overall power consumption of the array substrate.

(25) Referring to FIG. **1** and FIG. **2**, FIG. **1** is a top view of an array substrate according to some embodiments of the present disclosure, and FIG. **2** is a schematic structural diagram of film layers in the array substrate shown in FIG. **1** along a line A-A'. The array substrate **000** includes a display region **00a** and a non-display region **00b** located at the periphery of the display region **00a**. Here, the array substrate **000** includes a substrate **100**, and a first transistor **200** and a second transistor **300** that are disposed on the substrate **100**.

(26) The first transistor **200** in the array substrate **000** is disposed in the display region **00a**, and the first transistor **200** includes a first gate **201** and a first active layer **202**. The first gate **201** of the first transistor **200** is disposed on a side of the first active layer **202** facing away from the substrate **100**.

(27) The second transistor **300** in the array substrate **000** is disposed in the non-display region **00b**, and the second transistor **300** includes a second gate **301** and a second active layer **302**.

(28) In the present disclosure, the array substrate **000** further includes a data line D and a pixel electrode **400** that are disposed in the display region **00a**. The data line D is disposed on a side, close to the substrate **100**, of the first active layer **202** of the first transistor **200**, and the first active layer **202** of the first transistor **200** is lapped with the data line D. The pixel electrode **400** is disposed on a side, facing away from the substrate **100**, of the first gate **201** of the first transistor **200**, and the first active layer **202** of the first transistor **200** is lapped with the pixel electrode **400**. In this way, it is ensured that the first transistor **200** in the display region **00a** is electrically connected to the data line D, and is electrically connected to the pixel electrode **400**.

(29) In the embodiments of the present disclosure, the array substrate **000** further includes a gate line G disposed in the display region **00a**, and the gate line G is electrically connected to the gate **201** of the first transistor **200**.

(30) Exemplarily, a plurality of gate lines G and a plurality of data lines D are provided in the array substrate **000**. The plurality of gate lines G are arranged in parallel, the plurality of data lines D are arranged in parallel, and an extending direction of the gate line G is perpendicular to an extending direction of the data line D. In this way, every two adjacent gate lines G and every two adjacent data lines D form a sub-pixel region **00c**.

(31) A plurality of first transistors **200** are provided in the array substrate **000**, the plurality of first transistors **200** correspond to the plurality of sub-pixel regions **00c**, and each of the plurality of sub-pixel regions **00c** is provided with at least one corresponding first transistor **200**. Correspondingly, a plurality of pixel electrodes **400** are provided in the array substrate **000**, the plurality of pixel electrodes **400** correspond to the plurality of sub-pixel regions **00c**, and each of the plurality of sub-pixel regions **00c** is provided with a corresponding pixel electrode **400**. Moreover, the pixel electrode **400** in each sub-pixel region **00c** is electrically connected to the first active layer **202** of the first transistor **200** in the sub-pixel region **00c**.

(32) In this case, when a gate signal is loaded on the gate line G electrically connected to the first gate **201** of the first transistor **200** in a certain sub-pixel region **00c**, the first transistor **200** is in a turned-on state, that is, the first active layer **202** of the first transistor **200** can transmit an electrical signal, so that the data line D electrically connected to the first active layer **202** can apply a data signal to the pixel electrode **400** electrically connected to the first active layer **202**, and thus the pixel electrode **400** can load pixel voltage.

(33) In the present disclosure, a plurality of second transistors **300** are provided in the array substrate **000**, and the second transistors **300** are all transistors in a gate driver on array (GOA) circuit. Exemplarily, the array substrate **000** further includes a GOA circuit **00d** disposed in the non-display region **00b**. The GOA circuit **00d** is electrically connected to the gate line G in the display region **00a**. In this way, the GOA circuit **00d** can apply a gate signal to the gate line G, so that the GOA circuit **00d** can control the first transistor **200** to be turned on or turned off through the gate line G.

(34) In some embodiments, the first transistor **200** in the present disclosure is an oxide TFT, that is, the first active layer **202** of the first transistor **200** is made of an oxide semiconductor material. The second transistor **300** in the present disclosure is a low temperature poly-silicon (LTPS) TFT, that is, the second active layer **302** of the second transistor **300** is made of an LTPS semiconductor material. In this case, the array substrate **000** in the present disclosure is a low temperature polycrystalline oxide (LTPO) substrate. This array substrate **000** has the advantages of high mobility of LTPS and low leakage current of oxide. Because the oxide semiconductor material has high light transmittance, when the first transistor **200** in the display region **00a** is the oxide TFT, high light transmittance of the array substrate **000** can be ensured, thereby improving the display effect of the display panel integrated with the array substrate **000**. In addition, because the LTPS semiconductor material has the characteristic of high mobility, when the second transistor **300** in the non-display region **00b** is the LTPS TFT, it can be ensured that the second transistor **300** has a small size. Thus, the width of the non-display region **00b** of the array substrate **000** is small, thereby realizing the narrow bezel design of the display panel integrated with the array substrate **000**.

(35) In the embodiments of the present disclosure, the data line D in the array substrate **000** is disposed on the side, close to the substrate **100**, of the first active layer **202** of the first transistor **200**, the first gate **201** of the first transistor **200** in the array substrate **000** is disposed on the side, facing away from the substrate **100**, of the first active layer **202**, and the pixel electrode **400** in the array substrate **000** is disposed on the side, facing away from the substrate **100**, of the first gate **201** of the first transistor **200**. Therefore, it can be ensured that in the direction perpendicular to the substrate **100**, the vertical distance between the data line D and the pixel electrode **400** is relatively big, thereby ensuring that the parasitic capacitance generated between the data line D and the pixel electrode **400** is small. Thus, the probability that the parasitic capacitance interferes with the data signal loaded on the data line D is low, and the data signal loaded on the data line D can be normally transmitted to the pixel electrode **400**, thereby effectively reducing the power consumption of the array substrate **000**.

(36) In the present disclosure, the data line D in the array substrate **000** and the second gate **301** of the second transistor **300** are disposed in the same layer and made of the same material. It should

be noted that two structures being disposed in the same layer and made of the same material in the present disclosure means that the two structures are formed by one patterning process. For example, in the case that the data line D and the second gate **301** are disposed in the same layer and made of the same material, the data line D and the second gate **301** are formed by one patterning process. Here, one patterning process refers to thin film deposition, photoresist coating, exposure, development, etching and photoresist stripping. In this case, the manufacturing difficulty of the array substrate **000** can be effectively reduced on the premise of ensuring the small parasitic capacitance generated between the data line D and the pixel electrode **400**.

(37) In summary, the array substrate provided in the embodiments of the present disclosure includes a substrate, and a first transistor, a second transistor, a data line, and a pixel electrode that are disposed on the substrate. The data line in the array substrate is disposed on the side, close to the substrate, of the first active layer of the first transistor, the first gate of the first transistor in the array substrate is disposed on the side, facing away from the substrate, of the first active layer, and the pixel electrode in the array substrate is disposed on the side, facing away from the substrate, of the first gate of the first transistor. Therefore, it can be ensured that in the direction perpendicular to the substrate, the vertical distance between the data line and the pixel electrode is relatively big, thereby ensuring that the parasitic capacitance generated between the data line and the pixel electrode is small. Thus, the probability that the parasitic capacitor interferes with the data signal loaded on the data line is low, and the data signal loaded on the data line can be normally transmitted to the pixel electrode, thereby effectively reducing the power consumption of the array substrate. In addition, in the array substrate, the data line and the gate of the second transistor are disposed in the same layer and made of the same material, which can effectively reduce the manufacturing difficulty of the array substrate on the premise of ensuring the small parasitic capacitance generated between the data line and the pixel electrode.

(38) In the embodiments of the present disclosure, as shown in FIG. 3, FIG. 3 is a schematic structural diagram of film layers in another array substrate according to some embodiments of the present disclosure. The first active layer **202** of the first transistor **200** in the array substrate **000** includes a first conductor portion **202a** and a second conductor portion **202b** disposed opposite to each other, and a first channel region **202c** disposed between the first conductor portion **202a** and the second conductor portion **202b**. An orthographic projection of the first channel region **202c** of the first active layer **202** on the substrate **100** is within an orthographic projection of the first gate **201** on the substrate **100**.

(39) The orthographic projection of the data line D in the array substrate **000** on the substrate **100** has an overlapping region with the orthographic projection of the first conductor portion **202a** on the substrate **100**, and the data line D is lapped with the first conductor portion **202a** of the first active layer **202**. The orthographic projection of the pixel electrode **400** in the array substrate **000** on the substrate **100** has an overlapping region with the orthographic projection of the second conductor portion **202b** on the substrate **100**, and the pixel electrode **400** is lapped with the second conductor portion **202b** of the first active layer **202**.

(40) Therefore, when a gate drive signal is loaded on the first gate **201**, carriers are generated in the first channel region **202c** of the first active layer **202**, and the first conductor portion **202a** and the second conductor portion **202b** of the first active layer **202** can be electrically connected to each other, so that the corresponding data line D can transmit the data signal to the corresponding pixel electrode **400**. When loading the gate drive signal to the first gate **201** is stopped, carriers are no longer generated in the first channel region **202c** of the first active layer **202**, and the electrical connection between the first conductor portion **202a** and the second conductor portion **202b** of the first active layer **202** is cut off, so that the corresponding data line D stops transmitting the data signal to the corresponding pixel electrode **400**.

(41) In this case, when the data line D is electrically connected to the first conductor portion **202a** of the first active layer **202** in a direct lap joint manner, and the pixel electrode **400** is electrically

connected to the second conductor portion **202b** of the first active layer **202** in a direct lap joint manner, the electrical connection between the data line **D** and the first active layer **202** and the electrical connection between the pixel electrode **400** and the first active layer **202** can be achieved, without providing the source and the drain in the first transistor **200**. The first active layer **202** of the first transistor **200** is made of an oxide semiconductor material, which is a transparent semiconductor material, and the source and the drain of the TFT are generally made of an opaque metal material. Therefore, when the source and the drain are not provided in the first transistor **200**, the shielding of the first active layer **202** by the source and the drain can be reduced, thereby effectively increasing the aperture ratio of the array substrate **000**, and further improving the light transmittance of the array substrate **000**.

(42) In some embodiments, as shown in FIG. 3, the array substrate **000** further includes a transparent connecting electrode **500** disposed between the pixel electrode **400** and the first active layer **202** of the first transistor **200**. For example, the transparent connecting electrode **500** is disposed on the side, facing away from the substrate **100**, of the first gate **201** of the first transistor **200**. The orthographic projection of the transparent connecting electrode **500** on the substrate **100** has an overlapping region with the orthographic projection of the pixel electrode **400** on the substrate **100**, and has an overlapping region with the orthographic projection of the second conductor portion **202b** of the first active layer **202** on the substrate **100**. The pixel electrode **400** is lapped with the transparent connecting electrode **500**, and the transparent connecting electrode **500** is lapped with the second conductor portion **202b** of the first active layer **202**. In this way, the pixel electrode **400** is electrically connected to the second conductor portion **202b** of the first active layer **202** by means of the transparent connecting electrode **500**. Since the film layer between the pixel electrode **400** and the first active layer **202** in the array substrate **000** is relatively thick, the distance between the pixel electrode **400** and the first active layer **202** is also relatively long. Therefore, when the transparent connecting electrode **500** is disposed between the pixel electrode **400** and the first active layer **202**, and the pixel electrode **400** is lapped with the second conductor portion **202b** of the first active layer **202** by means of the transparent connecting electrode **500**, the risk of disconnection between the pixel electrode **400** and the second conductor portion **202b** of the first active layer **202** is low when they are electrically connected to each other. In addition, the transparent connecting electrode **500** is made of a transparent conductive material, which does not affect the aperture ratio of the array substrate **000**.

(43) In the embodiments of the present disclosure, as shown in FIG. 4, FIG. 4 is a schematic structural diagram of film layers in still another array substrate according to some embodiments of the present disclosure. The array substrate **000** further includes a light shielding layer **600**. The orthographic projection of the first channel region **202c** of the first active layer **202** in the first transistor **200** on the substrate **100** is within the orthographic projection of the light shielding layer **600** on the substrate **100**. In this way, light emitted to the first channel region **202c** of the first active layer **202** is shielded by the light shielding layer **600**, so as to prevent photon-generated carriers from generating in the first channel region **202c** under the irradiation of light, thereby avoiding the current leakage phenomenon of the first transistor **200**. It should be noted that, in other possible implementations, the light shielding layer **600** is not provided in products which do not require high electrical properties of the first transistor **200** and products in which the first active layer **201** of the first transistor **200** is made of a material with high optical stability, which can further improve the light transmittance of the array substrate **000**.

(44) In the present disclosure, in the array substrate **000**, the light shielding layer **600** and the data line **D** are disposed in the same layer and made of the same material. In this way, on the premise of ensuring that the light shielding layer **600** can normally shield the light emitted to the first channel region **202c** of the first active layer **202**, the manufacturing process steps of the array substrate **000** are simplified, and the costs are reduced.

(45) In some embodiments, the light shielding layer **600** in the array substrate **000** includes a

plurality of light shielding blocks (not marked in the drawings) in one-to-one correspondence with the plurality of first transistors **200**. The orthographic projection of the first channel region **202c** of the first active layer **202** in each first transistor **200** on the substrate **100** is within the orthographic projection of the corresponding light shielding block on the substrate **100**. In this way, it can be ensured that the light shielding layer **600** is only distributed at the position where the first channel region **202c** is provided in the array substrate **000**, and the light shielding layer **600** is not arranged at other positions, which ensures the high light transmittance of the array substrate **000**.

(46) In the embodiments of the present disclosure, there are a plurality of positional relationships between the first active layer **202** of the first transistor **200** and the second active layer **302** of the second transistor **300** in the array substrate **000**, and the following two optional implementations are taken as examples for illustration in the embodiments of the present disclosure.

(47) In a first optional implementation, as shown in FIG. 3 and FIG. 4, the first active layer **202** of the first transistor **200** is disposed in a different layer from the second active layer **302** of the second transistor **300**, and the second active layer **302** of the second transistor **300** is closer to the substrate **100** than the second gate **301** is. That is, the second transistor **300** is a top-gate TFT. It should be noted that two structures being disposed in different layers in the present disclosure means that the two structures are not in the same film layer, and other film layer structures exist between the two structures. For example, the first active layer **202** and the second active layer **302** being disposed in different layers means that other film layer structures exist between the first active layer **202** and the second active layer **302**.

(48) In the embodiments of the present disclosure, the second transistor **300** further includes a source **303** and a drain **304** which are lapped with the second active layer **302**. For example, the second active layer **302** includes a third conductor portion **302a** and a fourth conductor portion **302b** disposed opposite to each other, and a second channel region **302c** between the third conductor portion **302a** and the fourth conductor portion **302b**. The third conductor portion **302a** of the second active layer **302** is lapped with the source **303**, the fourth conductor portion **302b** of the second active layer **302** is lapped with the drain **304**, and the orthographic projection of the second channel region **302c** of the second active layer **302** on the substrate **100** is within the orthographic projection of the second gate **301** on the substrate **100**. It should be noted that the working principle of the second transistor **300** is similar to that of the first transistor **200**, and the working principle of the second transistor **300** is not repeated herein.

(49) In the present disclosure, the source **303** and the drain **304** of the second transistor **300** are disposed in the same layer and made of the same material, and the source **303** and the drain **304** of the second transistor **300** are both disposed on the side of the second gate **301** facing away from the substrate **100**.

(50) It should be noted that the film layer positions of the source **303** and the drain **304** of the second transistor **300** in the array substrate **000** have various cases.

(51) In a first case, as shown in FIG. 3 and FIG. 4, the source **303** and the drain **304** of the second transistor **300** are disposed in the same layer as the first active layer **202** of the first transistor **200**, but the material of the first active layer **202** is different from the material of the source **303** and the drain **304** of the second transistor **300**. It should be noted that two structures being disposed in the same layer but made of different materials in the present disclosure means that the two structures are disposed in the same film layer, but the materials are different. So, the two structures need to be formed in the same film layer by two patterning processes respectively. For example, the source **303** and the drain **304** of the second transistor **300** are usually made of a metal material, and the first active layer **202** of the first transistor **200** is made of an oxide semiconductor material.

(52) In this case, the source **303** and the drain **304** of the second transistor **300** need to be lapped with the second active layer **302** through a via hole in the array substrate **000**, and the first active layer **202** also needs to be lapped with the data line D through a via hole in the array substrate **000**. Therefore, in the case that the source **303** and the drain **304** of the second transistor **300** are

disposed in the same layer as the first active layer **202** of the first transistor **200**, different types of via holes are formed in the array substrate **000** by one patterning process before the source **303**, the drain **304** and the second active layer **302** are formed. In this way, the manufacturing difficulty of the array substrate **000** can be effectively reduced.

(53) Exemplarily, the array substrate **000** further includes a first gate insulating layer **001** disposed between the second gate **301** and the second active layer **302** of the second transistor **300**, and an auxiliary insulating layer **002** disposed between the first active layer **202** of the first transistor **200** and the data line **D**.

(54) The array substrate **000** is provided with a first via hole **V1** penetrating through the auxiliary insulating layer **002**, and a plurality of lap joint via holes **V2** penetrating through the auxiliary insulating layer **002** and the first gate insulating layer **001**. The first active layer **202** of the first transistor **200** is lapped with the data line **D** through the first via hole **V1**. The source **303** of the second transistor **300** is lapped with the second active layer **302** of the second transistor **300** through one lap joint via hole **V2**, and the drain **304** of the second transistor **300** is lapped with the second active layer **302** of the second transistor **300** through another lap joint via hole **V2**.

(55) In the array substrate **000**, the opening of the first via hole **V1** facing away from the substrate **100** and the openings of the lap joint via holes **V2** facing away from the substrate **100** are all on the surface of the auxiliary insulating layer **002** facing away from the substrate **100**.

(56) Therefore, in the manufacturing process of the array substrate **000**, after the auxiliary insulating layer **002** is manufactured, one patterning process is performed on the auxiliary insulating layer **002** to form the first via hole **V1** penetrating through the auxiliary insulating layer **002** and the plurality of lap joint via holes **V2** penetrating through the auxiliary insulating layer **002** and the first gate insulating layer **001** in the array substrate **000**. Subsequently, after the first active layer **202** of the first transistor **200** is formed, the first active layer **202** is lapped with the data line **D** through the first via hole **V1**; after the source **303** and the drain **304** of the second transistor **300** are formed, the source **303** is lapped with the second active layer **302** of the second transistor through one lap joint via hole **V2**, and the drain **304** is also lapped with the second active layer **302** of the second transistor through another lap joint via hole **V2**.

(57) In some embodiments, the array substrate **000** further includes a second gate insulating layer **003** and an interlayer dielectric layer **004**. The first gate insulating layer **001**, the auxiliary insulating layer **002**, the second gate insulating layer **003**, and the interlayer dielectric layer **004** in the array substrate **000** are stacked in a direction perpendicular to and going away from the substrate **100**.

(58) The second active layer **302** of the second transistor **300** is disposed on the side of the first gate insulating layer **001** close to the substrate **100**; the data line **D** and the second gate **301** of the second transistor **300** are both disposed between the first gate insulating layer **001** and the auxiliary insulating layer **002**; the first active layer **202** of the first transistor **200**, and the source **303** and the drain **304** of the second transistor **300** are all disposed between the auxiliary insulating layer **002** and the second gate insulating layer **003**; and the first gate **201** of the first transistor **200** is disposed between the second gate insulating layer **003** and the interlayer dielectric layer **004**.

(59) In a second case, as shown in FIG. 5, FIG. 5 is a schematic structural diagram of film layers in still another an array substrate according to some embodiments of the present disclosure. The source **303** and the drain **304** of the second transistor **300** are disposed in the same layer and made of the same material as the first gate **201** of the first transistor **200**. In this way, the source **303** and the drain **304** of the second transistor **300** which are made of the metal material and the first gate **201** of the first transistor **200** which is made of the metal material can be simultaneously formed by the same one patterning process, which effectively increases the utilization ratio of the metal material in the process of forming the array substrate **000**, thereby reducing the manufacturing cost of the array substrate **000**.

(60) In this case, the array substrate **000** also includes the first gate insulating layer **001**, the

auxiliary insulating layer **002**, the second gate insulating layer **003**, and the interlayer dielectric layer **004** which are stacked in the direction perpendicular to and going away from the substrate **100**.

(61) The second active layer **302** of the second transistor **300** is disposed on the side of the first gate insulating layer **001** close to the substrate **100**; the data line D and the second gate **301** of the second transistor **300** are both disposed between the first gate insulating layer **001** and the auxiliary insulating layer **002**; the first active layer **202** of the first transistor **200** is disposed between the auxiliary insulating layer **002** and the second gate insulating layer **003**; and the first gate **201** of the first transistor **200**, and the source **303** and the drain **304** of the second transistor **300** are all disposed between the second gate insulating layer **003** and the interlayer dielectric layer **004**.

(62) In addition, the array substrate **000** is further provided with a plurality of lap joint via holes V2 penetrating through the interlayer dielectric layer **004**, the second gate insulating layer **003**, the auxiliary insulating layer **002**, and the first gate insulating layer **001**. The source **303** of the second transistor **300** is lapped with the second active layer **302** of the second transistor **300** through one lap joint via hole V2, and the drain **304** of the second transistor **300** is lapped with the second active layer **302** of the second transistor **300** through another lap joint via hole V2.

(63) In a second optional implementation, as shown in FIG. 6, FIG. 6 is a schematic structural diagram of film layers in an array substrate according to some embodiments of the present disclosure. The first active layer **202** of the first transistor **200** and the second active layer **302** of the second transistor **300** are disposed in the same layer but are made of different materials, and the second gate **301** of the second transistor **300** is closer to the substrate **100** than the second active layer **302** is. That is, the second transistor **300** is a bottom-gate TFT.

(64) In the embodiments of the present disclosure, the second transistor **300** further includes a source **303** and a drain **304** which are lapped with the second active layer **302**. For the structure of the second transistor **300**, reference may be made to the descriptions in the foregoing embodiments, and details are not described herein again. The source **303** and the drain **304** of the second transistor **300** are disposed in the same layer and made of the same material as the first gate **201** of the first transistor **200**. In this way, the manufacturing difficulty of the array substrate **000** can be reduced, and the manufacturing cost of the array substrate **000** can be reduced.

(65) It should be noted that in the array substrate **000**, the data line D is lapped with the first active layer **202** of the first transistor **200** in various manners. The embodiments of the present disclosure are illustrated by taking the following two possible cases as examples.

(66) In a first possible case, as shown in FIG. 6, the array substrate **000** further includes a first gate insulating layer **001** disposed between the second gate **301** and the second active layer **302** of the second transistor **300**. Since the data line D is disposed in the same layer as the second gate **301**, and the second active layer **302** is disposed in the same layer as the first active layer **202**, the film layer between the data line D and the first active layer **202** is also the first gate insulating layer **001**. In this case, the array substrate **000** is provided with a first via hole V1 penetrating through the first gate insulating layer **001**, and the first active layer **202** of the first transistor **200** is lapped with the data line D through the first via hole V1.

(67) In some embodiments, the array substrate **000** further includes a second gate insulating layer **003** and an interlayer dielectric layer **004**. The first gate insulating layer **001**, the second gate insulating layer **003**, and the interlayer dielectric layer **004** are stacked in the direction perpendicular to and going away from the substrate **100**.

(68) The second gate **301** of the second transistor **300** and the data line D are both disposed on the side of the first gate insulating layer **001** close to the substrate **100**; the second active layer **302** of the second transistor **300** and the first active layer **202** of the first transistor **200** are both disposed between the first gate insulating layer **001** and the second gate insulating layer **003**; and the first gate **201** of the first transistor **200**, and the source **303** and the drain **304** of the second transistor **300** are all disposed between the second gate insulating layer **003** and the interlayer dielectric layer

004.

(69) In this case, the array substrate **000** is further provided with a plurality of lap joint via holes **V2** penetrating through the second gate insulating layer **003**. The source **303** of the second transistor **300** is lapped with the second active layer **302** of the second transistor **300** through one lap joint via hole **V2**, and the drain **304** of the second transistor **300** is lapped with the second active layer **302** of the second transistor **300** through another lap joint via hole **V2**.

(70) In a second possible case, as shown in FIG. 7, FIG. 7 is a schematic structural diagram of film layers in another array substrate according to some embodiments of the present disclosure. The array substrate **000** further includes a connecting electrode **700**. The connecting electrode **700** is lapped with the first active layer **202** of the first transistor **200** and the data line **D**. The connecting electrode **700**, the first gate **201** of the first transistor **200**, and the source **303** and the drain **304** of the second transistor **300** are disposed in the same layer and made of the same material.

(71) In this case, the source **303** and the drain **304** of the second transistor **300** need to be lapped with the second active layer **302** through the via holes in the array substrate **000**, and the connecting electrode **700** also needs to be lapped with the data line **D** and the first active layer **202** of the first transistor **200** through the via holes in the array substrate **000**. Therefore, in the case that the connecting electrode **700** is disposed in the same layer as the source **303** and the drain **304** of the second transistor **300**, different types of via holes are formed in the array substrate **000** by one patterning process before the source **303**, the drain **304** and the connecting electrode **700** are formed. In this way, the manufacturing difficulty of the array substrate **000** can be effectively reduced. In addition, because the connecting electrode **700**, the first gate **201** of the first transistor **200**, and the source **303** and the drain **304** of the second transistor **300** are disposed in the same layer and made of the same material, the first active layer **202** is electrically connected to the data line **D** by disposing the connecting electrode **700**, and the manufacturing difficulty of the array substrate **000** is not increased.

(72) Exemplarily, the array substrate **000** further includes a first gate insulating layer **001** disposed between the second gate **301** and the second active layer **302** of the second transistor **300**, and a second gate insulating layer **003** disposed between the first active layer **202** and the first gate **201** of the first transistor **200**.

(73) The array substrate **000** is provided with a first sub-via hole **V11** penetrating through the second gate insulating layer **003** and the first gate insulating layer **001**, and a second sub-via hole **V12** and a plurality of lap joint via holes **V2** penetrating through the second gate insulating layer **003**. The connecting electrode **700** is lapped with the first active layer **202** of the first transistor **200** through the second sub-via hole **V12**, and the connecting electrode **700** is lapped with the data line **D** through the first sub-via hole **V11**. In this way, the first active layer **202** is electrically connected to the data line **D** through the connecting electrode **700**. The source **303** of the second transistor **300** is lapped with the second active layer **302** of the second transistor **300** through one lap joint via hole **V2**, and the drain **304** of the second transistor **300** is lapped with the second active layer **302** of the second transistor **300** through another lap joint via hole **V2**.

(74) The opening of the first sub-via hole **V11** facing away from the substrate **100**, the opening of the second sub-via hole **V12** facing away from the substrate **100**, and the openings of the lap joint via holes **V2** facing away from the substrate **100** are all on the surface of the second gate insulating layer **003** facing away from the substrate.

(75) Therefore, in the manufacturing process of the array substrate **000**, after the second gate insulating layer **003** is manufactured, one patterning process is performed on the second gate insulating layer **003** to form the first sub-via hole **V11** penetrating through the second gate insulating layer **003** and the first gate insulating layer **001**, and the second sub-via hole **V12** and the plurality of lap joint via holes **V2** penetrating through the second gate insulating layer **003** in the array substrate **000**. Subsequently, after the connecting electrode **700**, and the source **303** and the drain **304** of the second transistor **300** are formed by one patterning process, the connecting

electrode **700** is lapped with the first active layer **202** through the second sub-via hole **V12** and is lapped with the data line **D** through the first sub-via hole **V11**, the source **303** is lapped with the second active layer **302** of the second transistor through one lap joint via hole **V2**, and the drain **304** is also lapped with the second active layer **302** of the second transistor through another lap joint via hole **V2**.

(76) In some embodiments, the array substrate **000** further includes an interlayer dielectric layer **004**. The first gate insulating layer **001**, the second gate insulating layer **003**, and the interlayer dielectric layer **004** in the array substrate **000** are stacked in the direction perpendicular to and going away from the substrate **100**.

(77) The second gate **301** of the second transistor **300** and the data line **D** are both disposed on the side of the first gate insulating layer **001** close to the substrate **100**; the second active layer **302** of the second transistor **300** and the first active layer **202** of the first transistor **200** are both disposed between the first gate insulating layer **001** and the second gate insulating layer **003**; and the first gate **201** of the first transistor **200**, and the source **303** and the drain **304** of the second transistor **300** are all disposed between the second gate insulating layer **003** and the interlayer dielectric layer **004**.

(78) It should be noted that in the second optional implementation, when the first active layer **202** of the first transistor **200** and the second active layer **302** of the second transistor **300** are disposed in the same layer but made of different materials, the overall thickness of the array substrate **000** can be reduced. For example, there is no need to provide the auxiliary insulating layer in the array substrate **000** in this case. In this way, it can be ensured that the overall stress in the array substrate **000** is low, which can reduce the probability of warpage at the edge of the array substrate **000**, thereby effectively improving the yield of the array substrate **000**.

(79) In the embodiments of the present disclosure, as shown in FIG. 3 to FIG. 7, the transparent connecting electrode **500** in the array substrate **000** is disposed on the side of the interlayer dielectric layer **004** facing away from the substrate **100**. The array substrate **000** further includes a planarization layer **005** disposed on the side of the transparent connecting electrode **500** facing away from the substrate **100**. The pixel electrode **400** in the array substrate **000** is disposed on the side of the planarization layer **005** facing away from the substrate **100**.

(80) The array substrate **000** is further provided with a connecting via hole **V3** penetrating through the planarization layer **005**, and a second via hole **V4** penetrating through the interlayer dielectric layer **004** and the second gate insulating layer **003**. The pixel electrode **400** is lapped with the transparent connecting electrode **500** through the connecting via hole **V3**, and the transparent connecting electrode **500** is lapped with the first active layer **202** of the first transistor **200** through the second via hole **V4**. For example, the transparent connecting electrode **500** is lapped with the second conductor portion **202b** of the first active layer **202** through the second via hole **V4**. In this way, it is ensured that the risk of disconnection between the pixel electrode **400** and the second conductor portion **202b** of the first active layer **202** is low when they are electrically connected to each other, without affecting the aperture ratio of the array substrate **000**.

(81) In some embodiments, as shown in FIG. 3 to FIG. 7, the array substrate **000** further includes a passivation layer **006** disposed on the side of the pixel electrode **400** facing away from the substrate **100**, and a common electrode layer **800** disposed on the side of the passivation layer **006** facing away from the substrate **100**. In this way, the pixel electrode **400** and the common electrode layer **800** can be integrated simultaneously in the array substrate **000**, thereby improving the integration level of the array substrate **000**. In a display panel integrated with the array substrate **000**, when an electric field is formed between the pixel electrode **400** and the common electrode layer **800** disposed in the array substrate **000**, the electric field can drive liquid crystals in the display panel to rotate, and the display panel can display a corresponding image.

(82) In the embodiments of the present disclosure, the common electrode layer **800** includes a light shielding portion **801** and a transparent conductive portion **802** disposed on the side of the light

shielding portion **801** facing away from the substrate **100**. The orthographic projection of the light shielding portion **801** of the common electrode layer **800** on the substrate **100** is not overlapped with the orthographic projection of the pixel electrode **400** on the substrate **100**. For example, the orthographic projection of the light shielding portion **801** of the common electrode layer **800** on the substrate **100** is in a mesh shape, and the orthographic projection of the light shielding portion **801** on the substrate **100** is between the orthographic projections of two adjacent pixel electrodes **400** on the substrate **100**. After the array substrate **000** is integrated in the display device, the light shielding portion **801** of the common electrode layer **800** can shield the light emitted to the adjacent sub-pixel region **00c** among the light emitted from a sub-pixel region **00c** in the array substrate **000**, thereby effectively reducing the probability of cross color occurring in the display device, and ensuring a good display effect of the display device.

(83) In the present disclosure, the orthographic projection of the data line D on the substrate **100** and the orthographic projection of the gate line G on the substrate **100** are within the orthographic projection of the light shielding portion **801** of the common electrode layer **800** on the substrate **100**. In addition, in the extending direction of the gate line G, the width of the light shielding portion **801** is greater than the width of the data line D; and in the extending direction of the data line D, the width of the light shielding portion **801** is greater than the width of the gate line G. Therefore, the light shielding portion **801** can shield the data line D and the gate line G, which can reduce the probability that the ambient light emitted to the array substrate is reflected by the data line D and the gate line G, thereby effectively reducing the probability that the array substrate reflects the ambient light.

(84) It should be noted that, when the light shielding portion **801** is disposed in the common electrode layer **800**, the light shielding portion **801** serves as a black matrix. Therefore, a black matrix is not provided in the color filter substrate of the display panel, which can effectively simplify the manufacturing process of the color filter substrate. Certainly, a black matrix may also be normally provided in the color filter substrate, which is not limited in the embodiments of the present disclosure.

(85) In some embodiments, the transparent conductive portion **802** of the common electrode layer **800** is a transparent planar electrode disposed on the entire layer. As such, the transparent conductive portion **802** covers the light shielding portion **801**. For example, the transparent conductive portion **802** of the common electrode layer **800** is made of a transparent conductive material such as indium zinc oxide (ITO) or indium tin oxide (ITO). In the present disclosure, the transparent conductive portion **802** of the common electrode layer **800** has a slit **803**. The light transmittance of the array substrate **000** is further improved by forming the slit in the transparent conductive portion **802**.

(86) In the embodiments of the present disclosure, the light shielding portion **801** of the common electrode layer **800** is made of a conductive material, and certainly can also be made of a non-conductive material. Exemplarily, when the light shielding portion **801** of the common electrode layer **800** is made of a conductive material, the conductive material is a metal material such as molybdenum, titanium, aluminum or an alloy. In this case, because the resistivity of these conductive materials is generally lower than the resistivity of the transparent conductive material, the resistance of the transparent conductive portion **802** of the common electrode layer **800** is reduced by means of the light shielding portion **801**, thereby ensuring that the potentials applied to various positions of the common electrode layer **800** are approximately the same. It should be noted that the light shielding portion **801** made of the metal material has certain light reflectivity. In order to reduce the reflectivity of the light shielding portion **801** to ambient light, the metal material is blackened. In this way, on the premise of ensuring that the light shielding portion **801** formed subsequently can normally shield light, the reflectivity of the light shielding portion **801** to ambient light is reduced. For example, when the metal material is molybdenum, after molybdenum is blackened, molybdenum becomes black molybdenum oxide. Molybdenum oxide also has certain

conductivity.

(87) When the light shielding portion **801** of the common electrode layer **800** is made of a non-conductive material, the non-conductive material is an organic material such as a black matrix material or an inorganic material such as a carbon material.

(88) In other possible implementations, the common electrode layer **800** only includes the transparent conductive portion **802**, that is, the common electrode layer **800** is a planar electrode disposed on the entire layer.

(89) In some embodiments, the array substrate **000** further includes a support part **900**. A portion of the support part **900** is disposed in the connecting via hole V3, and the other portion is disposed outside the connecting via hole V3. The portion of the support part **900** which is disposed outside the connecting via hole V3 is in contact with the color filter substrate to support the color filter substrate, so as to ensure a good stability of the process of aligning the array substrate **000** with the color filter substrate to form the display panel.

(90) In summary, the array substrate provided in the embodiments of the present disclosure includes a substrate, and a first transistor, a second transistor, a data line, and a pixel electrode that are disposed on the substrate. The data line in the array substrate is disposed on the side, close to the substrate, of the first active layer of the first transistor, the first gate of the first transistor in the array substrate is disposed on the side, facing away from the substrate, of the first active layer, and the pixel electrode in the array substrate is disposed on the side, facing away from the substrate, of the first gate of the first transistor. Therefore, it can be ensured that in the direction perpendicular to the substrate, the vertical distance between the data line and the pixel electrode is relatively big, thereby ensuring that the parasitic capacitance generated between the data line and the pixel electrode is small. Thus, the probability that the parasitic capacitor interferes with the data signal loaded on the data line is low, and the data signal loaded on the data line can be normally transmitted to the pixel electrode, thereby effectively reducing the power consumption of the array substrate. In addition, in the array substrate, the data line and the gate of the second transistor are disposed in the same layer and made of the same material, which can effectively reduce the manufacturing difficulty of the array substrate on the premise of ensuring the small parasitic capacitance generated between the data line and the pixel electrode.

(91) The embodiments of the present disclosure further provide a method of manufacturing an array substrate. The method of manufacturing the array substrate is used to manufacture the array substrate as shown in FIG. 1. The method of manufacturing the array substrate includes forming a first transistor, a second transistor, a data line, and a pixel electrode on a substrate.

(92) The array substrate includes a display region and a non-display region located at the periphery of the display region. The first transistor is disposed in the display region, and the second transistor is disposed in the non-display region. The first transistor includes a first gate and a first active layer, and the first gate is disposed on the side of the first active layer facing away from the substrate. The second transistor includes a second active layer and a second gate. The data line and the pixel electrode are both disposed in the display region. The data line is disposed on the side of the first active layer close to the substrate, and the data line is lapped with the first active layer; and the pixel electrode is disposed on the side of the first gate facing away from the substrate, and the pixel electrode is lapped with the first active layer. The data line and the second gate are disposed in the same layer and made of the same material.

(93) In summary, the method of manufacturing the array substrate provided in the embodiments of the present disclosure includes forming a first transistor, a second transistor, a data line and a pixel electrode on a substrate. The data line in the array substrate is disposed on the side, close to the substrate, of the first active layer of the first transistor, the first gate of the first transistor in the array substrate is disposed on the side, facing away from the substrate, of the first active layer, and the pixel electrode in the array substrate is disposed on the side, facing away from the substrate, of the first gate of the first transistor. Therefore, it can be ensured that in the direction perpendicular to

the substrate, the vertical distance between the data line and the pixel electrode is relatively big, thereby ensuring that the parasitic capacitance generated between the data line and the pixel electrode is small. Thus, the probability that the parasitic capacitor interferes with the data signal loaded on the data line is low, and the data signal loaded on the data line can be normally transmitted to the pixel electrode, thereby effectively reducing the power consumption of the array substrate. In addition, in the array substrate, the data line and the gate of the second transistor are disposed in the same layer and made of the same material, which can effectively reduce the manufacturing difficulty of the array substrate on the premise of ensuring the small parasitic capacitance generated between the data line and the pixel electrode.

(94) Referring to FIG. 8, FIG. 8 shows a method of manufacturing an array substrate according to some embodiments of the present disclosure. The method of manufacturing the array substrate is used to manufacture the array substrate shown in FIG. 3 or FIG. 4 in the above embodiments. The method of manufacturing the array substrate includes the following steps.

(95) In step S101, a second active layer is formed on the substrate.

(96) In some embodiments, the second active layer is made of an LTPS semiconductor material.

(97) Exemplarily, referring to FIG. 9, FIG. 9 is a schematic diagram of formation of a second active layer on a substrate according to some embodiments of the present disclosure. A thin film made of the LTPS semiconductor material is formed on a side of the substrate 100 by any one of deposition, coating, sputtering and the like, and then a one-time patterning process is performed on the thin film to form the second active layer 302. Here, the second active layer 302 is only disposed in the non-display region 00b.

(98) In step S102, a first gate insulating layer and a first conductive layer are sequentially formed on the second active layer.

(99) In some embodiments, the first gate insulating layer is made of silicon oxide, silicon nitride, or a mixture of silicon oxide and silicon nitride, and the thickness of the first gate insulating layer ranges from 600 Å to 2000 Å. The first conductive layer is made of a metal material such as aluminum, molybdenum or an alloy.

(100) Exemplarily, referring to FIG. 10, FIG. 10 is a schematic diagram of formation of the first gate insulating layer and the first conductive layer sequentially on the second active layer according to some embodiments of the present disclosure. The first gate insulating layer 001 and a first conductive thin film which is made of the metal material are sequentially formed on the side of the second active layer 302 facing away from the substrate 100 by any one of deposition, coating, sputtering and the like, and then the one-time patterning process is performed on the first conductive thin film to form the first conductive layer. Here, the first conductive layer includes a second gate 301 disposed in the non-display region 00b and a data line D disposed in the display region 00a. In other possible implementations, the first conductive layer further includes a light shielding layer 600 disposed in the display region 00a.

(101) When the first conductive layer only includes the second gate 301 and the data line D, the array substrate shown in FIG. 3 is manufactured. When the first conductive layer includes not only the second gate 301 and the data line D but also the light shielding layer 600, the array substrate shown in FIG. 4 is manufactured.

(102) In step S103, an auxiliary insulating layer is formed on the first conductive layer, and the auxiliary insulating layer is patterned.

(103) In some embodiments, the thickness of the auxiliary insulating layer ranges from 2000 Å to 8000 Å. The auxiliary insulating layer is a single-layered insulating layer, or a plurality of stacked insulating layers. When the auxiliary insulating layer is a single-layered insulating layer, the auxiliary insulating layer is made of silicon dioxide. When the auxiliary insulating layer is a plurality of stacked insulating layers, these insulating layers include an insulating layer made of silicon oxide, an insulating layer made of silicon nitride, an insulating layer made of silicon dioxide, and the like. In addition, it should be ensured that the insulating layer farthest from the

substrate **100** in the plurality of stacked insulating layers is an insulating layer made of silicon dioxide. In this case, since silicon dioxide has good compactness, it can be ensured that the first active layer made of the oxide semiconductor material subsequently formed on the auxiliary insulating layer is not conductorized, thereby ensuring the good electrical properties of the first transistor which is subsequently formed.

(104) Exemplarily, referring to FIG. **11**, FIG. **11** is a schematic diagram of formation of the auxiliary insulating layer on the first conductive layer according to some embodiments of the present disclosure. The auxiliary insulating layer **002** is formed on the side of the first conductive layer facing away from the substrate **100** by any one of deposition, coating, sputtering and the like, and then the one-time patterning process is performed on the auxiliary insulating layer **002** to form a first via hole **V1** and lap joint via holes **V2**. The first via hole **V1** is in the display region **00a**, and the first via hole **V1** penetrates through the auxiliary insulating layer **002**. The lap joint via holes **V2** are in the non-display region **00b**, and the lap joint via holes **V2** sequentially penetrate through the auxiliary insulating layer **002** and the first gate insulating layer **001**.

(105) In step **S104**, a second conductive layer is formed on the patterned auxiliary insulating layer.

(106) In some embodiments, the thickness of the second conductive layer ranges from 2000 Å to 8000 Å. The second conductive layer is made of a metal material such as aluminum, molybdenum or an alloy.

(107) Exemplarily, referring to FIG. **12**, FIG. **12** is a schematic diagram of formation of the second conductive layer according to some embodiments of the present disclosure. A second conductive thin film is formed on the side of the patterned auxiliary insulating layer **002** facing away from the substrate **100** by any one of deposition, coating, sputtering and the like, and then the one-time patterning process is performed on the second conductive thin film to form the second conductive layer. Here, the second conductive layer is only disposed in the non-display region **00b**, and the second conductive layer includes a source **303** and a drain **304**. The source **303** and the drain **304** are both lapped with the second active layer **302** through the lap joint via holes **V2**. In this way, the source **303**, the drain **304**, the second gate **301** and the second active layer **302** form a second transistor **300** disposed in the non-display region **00b**.

(108) In step **S105**, a first active layer is formed on the patterned auxiliary insulating layer.

(109) In some embodiments, the thickness of the first active layer ranges from 150 Å to 1500 Å. The first active layer is made of an oxide semiconductor material.

(110) Exemplarily, referring to FIG. **13**, FIG. **13** is a schematic diagram of formation of the first active layer according to some embodiments of the present disclosure. A thin film made of the oxide semiconductor material is formed on the side of the patterned auxiliary insulating layer **002** facing away from the substrate **100** by any one of deposition, coating, sputtering, and the like, and then the one-time patterning process is performed on the thin film to form the first active layer **202**. Here, the first active layer **202** is only disposed in the display region **00a**, and the first active layer **202** is lapped with the data line **D** through the first via hole **V1**.

(111) In step **S106**, a second gate insulating layer, a third conductive layer, an interlayer dielectric layer, a transparent connecting electrode, a planarization layer, a pixel electrode, a passivation layer, a support part and a common electrode layer are sequentially formed on the first active layer.

(112) In the embodiments of the present disclosure, the second gate insulating layer, the third conductive layer, the interlayer dielectric layer, the transparent connecting electrode, the planarization layer, the pixel electrode, the passivation layer, the support part and the common electrode layer are sequentially formed on the substrate on which the first active layer is formed. The second gate insulating layer is made of silicon dioxide, which can prevent the first active layer made of the oxide semiconductor material from being conductorized. The third conductive layer includes a first gate disposed in the display region. The first gate and the first active layer form a first transistor disposed in the display region. In this way, the array substrate **000** shown in FIG. **3** or FIG. **4** is formed. It should be noted that these preparation processes are conventional processes,

and thus are not repeated herein.

(113) Referring to FIG. 14, FIG. 14 shows a method of manufacturing an array substrate according to some embodiments of the present disclosure. The method of manufacturing the array substrate is used to manufacture the array substrate shown in FIG. 7 in the above embodiments. The method of manufacturing the array substrate includes the following steps.

(114) In step S201, a first conductive layer is formed on a substrate.

(115) In some embodiments, the first conductive layer is made of a metal material such as aluminum, molybdenum or an alloy.

(116) Exemplarily, referring to FIG. 15, FIG. 15 is a schematic diagram of formation of the first conductive layer according to some embodiment of the present disclosure. A first conductive thin film made of the metal material is formed on a side of the substrate 100 by any one of deposition, coating, sputtering and the like, and then the one-time patterning process is performed on the first conductive thin film to form the first conductive layer. Here, the first conductive layer includes a second gate 301 disposed in the non-display region 00b, and a data line D disposed in the display region 00a.

(117) In step S202, a first gate insulating layer is formed on the first conductive layer.

(118) In some embodiments, the thickness of the first gate insulating layer ranges from 1000 Å to 6000 Å. The first gate insulating layer is a single-layered insulating layer, or a plurality of stacked insulating layers. When the first gate insulating layer is a single-layered insulating layer, the first gate insulating layer is made of silicon dioxide. When the first gate insulating layer is a plurality of stacked insulating layers, these insulating layers include an insulating layer made of silicon oxide, an insulating layer made of silicon nitride, an insulating layer made of silicon dioxide, and the like. In addition, it should be ensured that the insulating layer farthest from the substrate 100 in the plurality of stacked insulating layers is an insulating layer made of silicon dioxide. In this case, since silicon dioxide has good compactness, it can be ensured that the first active layer made of the oxide semiconductor material subsequently formed on the first gate insulating layer is not conductorized, thereby ensuring the good electrical properties of the first transistor which is subsequently formed.

(119) Exemplarily, referring to FIG. 16, FIG. 16 is a schematic diagram of formation the first gate insulating layer on the first conductive layer according to some embodiments of the present disclosure. The first gate insulating layer 001 is formed on the side of the first conductive layer facing away from the substrate 100 by any one of deposition, coating, sputtering, and the like.

(120) In step S203, a second active layer and a first active layer are respectively formed on the first gate insulating layer.

(121) In some embodiments, the second active layer is made of an LTPS semiconductor material, and the first active layer is made of an oxide semiconductor material.

(122) Exemplarily, referring to FIG. 17, FIG. 17 is a schematic diagram of formation of the second active layer on the first gate insulating layer according to some embodiment of the present disclosure. A thin film made of the LTPS semiconductor material is formed on the side of the first gate insulating layer 001 facing away from the substrate 100 by any one of deposition, coating, sputtering and the like, and then the one-time patterning process is performed on the thin film to form the second active layer 302. Here, the second active layer 302 is only disposed in the non-display region 00b.

(123) Referring to FIG. 18, FIG. 18 is a schematic diagram of formation of the first active layer on the first gate insulating layer according to some embodiments of the present disclosure. A thin film made of the oxide semiconductor material is formed on the side of the first gate insulating layer 001 facing away from the substrate 100 by any one of deposition, coating, sputtering and the like, and then the one-time patterning process is performed on the thin film to form the first active layer 202. Here, the first active layer 202 is disposed in the display region 00a.

(124) In step S204, a second gate insulating layer is formed on the first active layer and the second

active layer, and the second gate insulating layer is patterned.

(125) In some embodiments, the second gate insulating layer is made of silicon oxide, silicon nitride, or a mixture of silicon oxide and silicon nitride.

(126) Exemplarily, referring to FIG. 19, FIG. 19 is a schematic diagram of formation of the second gate insulating layer on the first active layer and the second active layer according to some embodiments of the present disclosure. The second gate insulating layer **003** is formed on the side, facing away from the substrate **100**, of the first active layer **202** and the second active layer **302** by any one of deposition, coating, sputtering, and the like, and then the one-time patterning process is performed on the second gate insulating layer **003** to form lap joint via holes **V2**, a first sub-via hole **V11**, and a second sub-via hole **V12**. The lap joint via holes **V2** are in the non-display region **00b**, and the lap joint via holes **V2** penetrate through the second gate insulating layer **003**. The first sub-via hole **V11** and the second sub-via hole **V12** are in the display region **00a**, the first sub-via hole **V11** sequentially penetrates through the second gate insulating layer **003** and the first gate insulating layer **001**, and the second sub-via hole **V12** penetrates through the second gate insulating layer **003**.

(127) In step **S205**, a second conductive layer is formed on the second gate insulating layer.

(128) In some embodiments, the second conductive layer is made of a metal material such as aluminum, molybdenum or an alloy.

(129) Exemplarily, referring to FIG. 20, FIG. 20 is a schematic diagram of formation of the second conductive layer on the second gate insulating layer according to some embodiment of the present disclosure. A second conductive thin film is formed on the side of the patterned second gate insulating layer **003** facing away from the substrate **100** by any one of deposition, coating, sputtering and the like, and then the one-time patterning process is performed on the second conductive thin film to form the second conductive layer. Here, the second conductive layer includes a source **303** and a drain **304** that are disposed in the non-display region **00b**, and a connecting electrode **700** and a first gate **201** that are disposed in the display region **00a**. The source **303** and the drain **304** are lapped with the second active layer **302** through the lap joint via holes **V2**. In this way, the source **303**, the drain **304**, the second gate **301**, and the second active layer **302** form the second transistor **300** disposed in the non-display region **00b**. The first gate **201** is insulated from the first active layer **202**, and the first gate **201** and the first active layer **202** form the first transistor **200** disposed in the display region **00a**. The connecting electrode **700** is lapped with the first active layer **202** and the data line **D** through the second sub-via hole **V12** and the first sub-via hole **V11** respectively. In this way, the first active layer **202** is lapped with the data line **D** through the connecting electrode **700**.

(130) In step **S206**, an interlayer dielectric layer, a transparent connecting electrode, a planarization layer, a pixel electrode, a passivation layer, a support part and a common electrode layer are sequentially formed on the second conductive layer.

(131) In the embodiments of the present disclosure, the interlayer dielectric layer, the transparent connecting electrode, the planarization layer, the pixel electrode, the passivation layer, the support part and the common electrode layer are sequentially formed on the substrate on which the second conductive layer is formed. In this way, the array substrate **000** shown in FIG. 7 can be formed. It should be noted that the preparation processes are conventional processes, and thus are not repeated herein.

(132) It should be further noted that the one-time patterning process in the above embodiments includes photoresist coating, exposure, development, etching and photoresist stripping. For the working principle of the array substrate prepared in the above embodiments, reference may be made to the embodiments described above in respect of the structure of the array substrate, and details are not repeated herein.

(133) The embodiments of the present disclosure further provide a display panel. The display panel includes an array substrate and a color filter substrate disposed opposite to each other, and a liquid

crystal layer disposed between the array substrate and the color filter substrate. The array substrate is the array substrate shown in FIG. 1, FIG. 3, FIG. 4, FIG. 5, FIG. 6, or FIG. 7.

(134) The embodiments of the present disclosure further provide a display device. The display device may be any product or component having a display function such as a mobile phone, a tablet computer, a television, a display, a notebook computer, a digital photo frame, or a navigator. The display device includes the display panel in the above embodiments and a backlight source. The backlight source is disposed on the side of the array substrate facing away from the color filter substrate.

(135) It should be noted that in the drawings, the dimensions of layers and regions may be exaggerated for clarity of illustration. Also, it is understood that when an element or layer is referred to as being “on” another element or layer, it may be directly on the other element or there may be an intervening layer. In addition, it is understood that when an element or layer is referred to as being “under” another element or layer, it may be directly under the other element or there may be more than one intervening layer or element. In addition, it is also understood that when a layer or element is referred to as being “between” two layers or elements, it may be the only layer between two layers or elements, or there may be more than one intervening layer or element. Similar reference numerals indicate similar elements throughout.

(136) In the present disclosure, the terms “first” and “second” are used for descriptive purposes only and are not to be construed as indicating or implying relative importance. The term “a plurality of” refers to two or more, unless specifically defined otherwise.

(137) The descriptions above are merely optional embodiments of the present disclosure, and are not intended to limit the present disclosure. Any modifications, equivalent substitutions, improvements, and the like made within the spirit and principles of the present disclosure shall be included within the protection scope of the present disclosure.

Claims

1. An array substrate, comprising a display region and a non-display region located at a periphery of the display region, the array substrate comprising: a substrate; a first transistor and a second transistor that are disposed on the substrate, wherein the first transistor is disposed in the display region, and the second transistor is disposed in the non-display region; the first transistor comprises a first gate and a first active layer, the first gate being disposed on a side of the first active layer facing away from the substrate; and the second transistor comprises a second active layer and a second gate; and a data line and a pixel electrode that are disposed in the display region, wherein the data line is disposed on a side of the first active layer close to the substrate and is lapped with the first active layer, and the pixel electrode is disposed on a side of the first gate facing away from the substrate and is lapped with the first gate; wherein the data line and the second gate are disposed in a same layer and made of a same material.

2. The array substrate according to claim 1, wherein the first active layer comprises a first conductor portion and a second conductor portion opposite to each other, and a first channel region disposed between the first conductor portion and the second conductor portion, an orthographic projection of the first channel region on the substrate is within an orthographic projection of the first gate on the substrate; wherein an orthographic projection of the data line on the substrate has an overlapping region with an orthographic projection of the first conductor portion on the substrate, and the first conductor portion is lapped with the data line; and an orthographic projection of the pixel electrode on the substrate has an overlapping region with an orthographic projection of the second conductor portion on the substrate, and the pixel electrode is lapped with the second conductor portion.

3. The array substrate according to claim 2, further comprising: a transparent connecting electrode disposed between the pixel electrode and the first active layer, wherein an orthographic projection

- of the transparent connecting electrode on the substrate has an overlapping region with the orthographic projection of the pixel electrode on the substrate, and has an overlapping region with the orthographic projection of the second conductor portion on the substrate, and the pixel electrode is lapped with the second conductor portion by means of the transparent connecting electrode.
4. The array substrate according to claim 2, further comprising: a light shielding layer, wherein the light shielding layer and the data line are disposed in a same layer and made of a same material, and the orthographic projection of the first channel region on the substrate is within an orthographic projection of the light shielding layer on the substrate.
5. The array substrate according to claim 4, wherein a plurality of first transistors are provided, and the light shielding layer comprises a plurality of light shielding blocks in one-to-one correspondence with the plurality of first transistors, wherein the orthographic projection of the first channel region of the first active layer in the first transistor on the substrate is within an orthographic projection of the corresponding light shielding block on the substrate.
6. The array substrate according to claim 1, wherein the first active layer and the second active layer are disposed in different layers, and the second active layer is closer to the substrate than the second gate is.
7. The array substrate according to claim 6, wherein the second transistor further comprises: a source and a drain which are lapped with the second active layer, wherein the source and the drain are disposed in a same layer and made of a same material, and the source and the drain are both disposed on a side of the second gate facing away from the substrate; wherein the source, the drain and the first active layer are disposed in the same layer, and a material of the first active layer is different from a material of the source and the drain; or the source, the drain and the first gate are disposed in the same layer and made of the same material.
8. The array substrate according to claim 7, wherein in a case that the source, the drain and the first active layer are disposed in the same layer, the array substrate further comprises: a first gate insulating layer disposed between the second gate and the second active layer, and an auxiliary insulating layer disposed between the first active layer and the data line; the array substrate is provided with a first via hole penetrating through the auxiliary insulating layer, and a plurality of lap joint via holes penetrating through the auxiliary insulating layer and the first gate insulating layer, wherein the first active layer is lapped with the data line through the first via hole, the source is lapped with the second active layer through one of the lap joint via holes, and the drain is lapped with the second active layer through another one of the lap joint via holes; wherein an opening of the first via hole facing away from the substrate and openings of the lap joint via holes facing away from the substrate are all on a surface of the auxiliary insulating layer facing away from the substrate.
9. The array substrate according to claim 8, further comprising: a second gate insulating layer and an interlayer dielectric layer; wherein the second active layer is disposed on a side of the first gate insulating layer close to the substrate; the data line and the second gate are both disposed between the first gate insulating layer and the auxiliary insulating layer; the first active layer, the source and the drain are all disposed between the auxiliary insulating layer and the second gate insulating layer; and the first gate is disposed between the second gate insulating layer and the interlayer dielectric layer.
10. The array substrate according to claim 1, wherein the first active layer and the second active layer are disposed in a same layer but made of different materials, and the second gate is closer to the substrate than the second active layer is.
11. The array substrate according to claim 10, wherein the second transistor further comprises: a source and a drain which are lapped with the second active layer, wherein the source and the drain are disposed in a same layer and made of a same material, and the source and the drain are both disposed on a side of the second active layer facing away from the substrate; wherein the array

substrate further comprises: a connecting electrode, wherein the connecting electrode is lapped with the first active layer and the data line, and the connecting electrode, the first gate, the source and the drain are disposed in a same layer and made of a same material.

12. The array substrate according to claim 11, further comprising: a first gate insulating layer disposed between the second gate and the second active layer, and a second gate insulating layer disposed between the first active layer and the first gate; wherein the array substrate is provided with a first sub-via hole penetrating through the second gate insulating layer and the first gate insulating layer, and a second sub-via hole and a plurality of lap joint via holes penetrating through the second gate insulating layer, wherein the connecting electrode is lapped with the first active layer through the second sub-via hole, and is lapped with the data line through the first sub-via hole, the source is lapped with the second active layer through one of the lap joint via holes, and the drain is lapped with the second active layer through another one of the lap joint via holes; wherein an opening of the first sub-via hole facing away from the substrate, an opening of the second sub-via hole facing away from the substrate, and openings of the lap joint via holes facing away from the substrate are all on a surface of the second gate insulating layer facing away from the substrate.

13. The array substrate according to claim 12, further comprising: an interlayer dielectric layer; wherein the second gate and the data line are disposed on a side of the first gate insulating layer close to the substrate; the first active layer and the second active layer are disposed between the first gate insulating layer and the second gate insulating layer; and the source, the drain, the connecting electrode and the first gate are all disposed between the second gate insulating layer and the interlayer dielectric layer.

14. The array substrate according to claim 9, further comprising: a transparent connecting electrode disposed on a side of the interlayer dielectric layer facing away from the substrate, and a planarization layer disposed on a side of the transparent connecting electrode facing away from the substrate, the pixel electrode being disposed on a side of the planarization layer facing away from the substrate; wherein the array substrate is provided with a connecting via hole penetrating through the planarization layer, and a second via hole penetrating through the interlayer dielectric layer and the second gate insulating layer, the pixel electrode is lapped with the transparent connecting electrode through the connecting via hole, and the transparent connecting electrode is lapped with the first active layer through the second via hole.

15. The array substrate according to claim 14, further comprising: a passivation layer disposed on a side of the pixel electrode facing away from the substrate, and a common electrode layer disposed on a side of the passivation layer facing away from the substrate.

16. The array substrate according to claim 15, wherein the common electrode layer comprises: a light shielding portion and a transparent conductive portion disposed on a side of the light shielding portion facing away from the substrate, wherein an orthographic projection of the light shielding portion on the substrate is not overlapped with an orthographic projection of the pixel electrode on the substrate.

17. The array substrate according to claim 14, further comprising: a support part, wherein a portion of the support part is in the connecting via hole, and the other portion is outside the connecting via hole.

18. The array substrate according to claim 15, wherein the first transistor is an oxide thin film transistor, and the second transistor is a low temperature poly-silicon thin film transistor.

19. The array substrate according to claim 18, further comprising: a gate drive circuit disposed in the non-display region, wherein the gate drive circuit comprises a plurality of second transistors.

20. A display panel, comprising: an array substrate and a color filter substrate disposed opposite to each other, and a liquid crystal layer disposed between the array substrate and the color filter substrate, wherein the array substrate comprises a display region and a non-display region located at a periphery of the display region, the array substrate comprising: a substrate; a first transistor and

a second transistor that are disposed on the substrate, wherein the first transistor is disposed in the display region, and the second transistor is disposed in the non-display region: the first transistor comprises a first gate and a first active layer, the first gate being disposed on a side of the first active layer facing away from the substrate; and the second transistor comprises a second active layer and a second gate; and a data line and a pixel electrode that are disposed in the display region, wherein the data line is disposed on a side of the first active layer close to the substrate and is lapped with the first active layer, and the pixel electrode is disposed on a side of the first gate facing away from the substrate and is lapped with the first active layer; wherein the data line and the second gate are disposed in a same layer and made of a same material.
